Figure 1

The Marie Short House



Taken by Unknown Author

How did Glenn Murcutt revolutionize vernacular architecture in conjunction with modern design in rural New South Wales, Australia through the use of the Marie Short House Project?

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Perched on stilts approximately 3 feet above the ground, the Marie Short House or otherwise known as ‘The Kempsey House’ is situated in a rural expanse along the Maria River in Kempsey, located north of Sydney, Australia. Built predominantly from locally-sourced timber using traditional post-and-beam techniques reminiscent of those employed in Australian woolsheds, the dwelling exudes the rustic appeal commonly associated with farm structures in the region. Its architectural design, reminiscent of the typical local, Australian architectural tradition seamlessly melds with its countryside backdrop, earning it recognition as an exemplary embodiment of vernacular Architecture *(Craven , 2019)*.

Marie Short, the mother of one of Murcutt's former clients commissioned Murcutt to design an extension for a farmhouse *(The Marie Short House)*, which he then later purchased off Marie. While evaluating the renovation costs before initiating construction, Murcutt hinted to Marie that it could be worthwhile exploring the option of designing a brand-new house, as the difference in cost may be more feasible. The Marie Short House now stands, as a revered architectural emblem, acclaimed both locally and globally *(Butler , 2022)*.

My case study explores how Glenn Murcutt's design strategies in the Marie Short House Project effectively respond to the local environment, climate, and cultural context of rural New South Wales, Australia. It focuses on Murcutt's innovative fusion of modern design principles with traditional vernacular architecture, showcasing his ability to push

the boundaries of sustainable and contextually sensitive architectural practices. Specifically, this study examines how Murcutt integrates elements of modern design, such as clean lines, minimalist aesthetics, and innovative building materials, to create a harmonious blend of tradition and innovation in his architectural approach.

Firstly, Glenn Murcutt, an architect hailing from Australia, integrates modernism, naturalism and environmentalism into his work. He operates autonomously throughout the entire process, delicately adjusting his designs to fit the land and weather conditions. Murcutt utilizes various materials like metal, wood and stone, with a keen awareness of their environmental impact. Incorporating natural elements such as light, water, wind and sun, he ensures that his designs blend seamlessly with their environment and effectively respond to ecological considerations *(Silvetti , 2002)*.

The Marie Short House design is straightforward, featuring two similar pavilions (as seen in *Figure 1)* —one for sleeping and one for living. Each pavilion has six timber bays, with the last two bays forming an open porch. A thickened wall between them creates a corridor for rainwater collection. Inside, the kitchen and bathrooms are designed with individualized components grouped together. The living pavilion faces north to maximize sunlight, with retractable metal louvers for light and privacy control, promoting energy efficiency. The roof is designed with curved corrugated metal sheets for ventilation, enhancing airflow and reducing the need for artificial cooling. This design approach not only enhances environmental sustainability but also creates a

comfortable living space attuned to its natural surroundings *(Heneghan, Gusheh, Lassen, Seyama., 2008)*. Constructing the Marie-Short House, a pivotal milestone in Murcutt's career, ultimately led him to develop his unique architectural style.

The central connection between the two offset pavilions is highlighted with pivoting doors, creating a seamless transition between the communal areas. This arrangement enhances connectivity rather than just serving as a passageway. Upon entering the house, one immediately grasps the concepts of prospect and refuge: while the surrounding farmland is visible from most rooms, the adaptable triple-skin facade effectively moderates the climatic conditions *(Butler , 2022)*.

In the case of the Marie Short House, Murcutt employed detailed computer-aided modeling to comprehensively assess the impact of natural ventilation, shading devices, and reinterpreted vernacular elements on the house's performance and spatial quality. For the performance analysis, simulations utilized Brisbane's weather data as it was the closest available. The warmest day of the year, typically on the 1st of January, was chosen to assess the impact of Murcutt's environmental design principles on the house's comfort conditions. Murcutt's deep comprehension of the site's landscape, views, and climate conditions guided him in creating a house primarily aimed at providing shelter, with internal spaces adaptable to suit the needs of its occupants *(Lecaro et al., 2017)*.

The simplicity of the building, its ability to accommodate evolving requirements, and its sensitivity to climate considerations all directly address the urgent issues surrounding the environmental crisis aiding future generations *(Butler , 2022)*.

The indigenous people of Western Australia express a sentiment: "to touch this earth lightly," advocating for minimal disturbance of nature. Murcutt's architectural approach reflects this philosophy, with houses elevated on stilts or supported by footings that minimize disruption to the land *(Engelen , 2011)*.

In conclusion, Glenn Murcutt's transformative influence on vernacular architecture in rural New South Wales stems from his adept fusion of modern design principles with traditional building practices and environmental awareness. Illustrated through projects like the Marie Short House, Murcutt showcases a profound grasp of the local landscape, climate, and cultural context. Utilizing locally sourced materials and pioneering construction techniques, he crafts buildings that seamlessly integrate with their surroundings while pushing the boundaries of sustainability and contextually sensitive design. Murcutt's architectural creations, characterized by their simplicity, flexibility, and environmental mindfulness, epitomize a holistic approach that addresses the urgent challenges posed by the environmental crisis. By blending tradition with innovation and

prioritizing ecological responsibility, Murcutt has significantly reshaped rural architecture in New South Wales and beyond.

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